STANDING ORDER PROCESSOR

COBOL Program

Aim

To write a COBOL program that processes standing orders by validating account numbers, checking transaction amounts and available balances, updating balances for successful orders, and displaying the total amount processed successfully.

Algorithm

1. Read the number of accounts.
2. Read each account number and its balance.
3. Read the number of standing orders.
4. Read each order ID, account number, and order amount.
5. For each order, search for the corresponding account.
6. If the account does not exist, reject the order as an unknown account.
7. If the amount is zero or negative, reject the order as an invalid amount.
8. If the balance is insufficient, reject the order.
9. Otherwise, subtract the order amount from the account balance, add it to the successful total, and display success.
10. Display the total amount processed successfully and stop.

COBOL Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. STANDING-ORDER.

DATA DIVISION.
WORKING-STORAGE SECTION.

01 ACCOUNT-COUNT PIC 9(3).
01 ORDER-COUNT PIC 9(4).
01 I PIC 9(4) VALUE 0.
01 J PIC 9(4) VALUE 0.

01 ACCOUNT-TABLE.
   05 ACCOUNT-ENTRY OCCURS 100 TIMES.
      10 ACCOUNT-ID PIC X(2).
      10 BALANCE PIC 9(9) VALUE 0.

01 ORDER-TABLE.
   05 ORDER-ENTRY OCCURS 5000 TIMES.
      10 ORDER-ID PIC X(2).
      10 ORDER-ACCOUNT PIC X(10).
      10 ORDER-AMOUNT PIC S9(9).

01 FOUND-FLAG PIC X VALUE 'N'.
01 TOTAL-SUCCESS PIC 9(12) VALUE 0.
01 TOTAL-OUTPUT PIC Z(11)9.

PROCEDURE DIVISION.
    ACCEPT ACCOUNT-COUNT

    PERFORM VARYING I FROM 1 BY 1
        UNTIL I > ACCOUNT-COUNT
```

```
    ACCEPT ACCOUNT-ID(I)
    ACCEPT BALANCE(I)
END-PERFORM

ACCEPT ORDER-COUNT

PERFORM VARYING I FROM 1 BY 1
    UNTIL I > ORDER-COUNT
    ACCEPT ORDER-ID(I)
    ACCEPT ORDER-ACCOUNT(I)
    ACCEPT ORDER-AMOUNT(I)
END-PERFORM

PERFORM VARYING I FROM 1 BY 1
    UNTIL I > ORDER-COUNT
    MOVE 'N' TO FOUND-FLAG

    PERFORM VARYING J FROM 1 BY 1
        UNTIL J > ACCOUNT-COUNT
        IF ORDER-ACCOUNT(I) = ACCOUNT-ID(J)
            MOVE 'Y' TO FOUND-FLAG
            EXIT PERFORM
        END-IF
    END-PERFORM

    IF FOUND-FLAG = 'N'
        DISPLAY ORDER-ID(I)
            ' REJECTED: UNKNOWN ACCOUNT'
    ELSE
        IF ORDER-AMOUNT(I) <= 0
            DISPLAY ORDER-ID(I)
                ' REJECTED: INVALID AMOUNT'
        ELSE
            IF BALANCE(J) >= ORDER-AMOUNT(I)
                SUBTRACT ORDER-AMOUNT(I)
                    FROM BALANCE(J)
                ADD ORDER-AMOUNT(I) TO TOTAL-SUCCESS
                DISPLAY ORDER-ID(I) ' SUCCESS'
            ELSE
                DISPLAY ORDER-ID(I)
                    ' REJECTED: INSUFFICIENT FUNDS'
            END-IF
        END-IF
    END-IF
END-PERFORM

MOVE TOTAL-SUCCESS TO TOTAL-OUTPUT
DISPLAY 'Successful total: '
    FUNCTION TRIM(TOTAL-OUTPUT)

STOP RUN.
```

Sample Input

```
2
A1
1000
A2
300
3
01
A1
200
02
A2
500
03
A1
400
```

Expected Output

```
01 SUCCESS
02 REJECTED: INSUFFICIENT FUNDS
03 SUCCESS
Successful total: 600
```

Result

The COBOL program validates and processes standing orders, updates account balances for successful transactions, rejects invalid or unaffordable orders, and displays the successful total.

Note: Each ACCEPT statement reads one input field, so enter each value on a separate line when using this sample input.